

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 2: Structural Properties of Preferences and Utility Functions

Notes on the Chapter

This chapter has three pieces:

1. It introduces monotonicity and convexity properties of preferences, and connects these properties and continuity of preferences to utility representations.
2. It relates these properties to the standard two-dimensional indifference curve diagram.
3. It introduces separability, quasi-linearity, and homotheticity of preferences, and corresponding properties of utility functions.

Accompanying the chapter are appendices on simple real analysis (stressing continuity) and convexity.

This is, for the most part, a relatively easy chapter, at least for students who understand continuity and convexity in mathematical terms. Students who lack this background should be strongly encouraged to spend as much time on the appendices as on the chapter itself, and even students with a good background in these matters should review the appendices carefully.

I stress the point that while some properties of utility functions (quasi-concavity, for example) hold for all representations of given preferences if they hold for any one, others (for instance, continuity) do not. Hence, we have results taking the form *If preferences are convex, every representation is quasi-concave* while other results read *If preferences are continuous, some representation can be found that is continuous*.

In a couple of places, the chapter can bog down (taking students down with it). First, there is the proof of Debreu's theorem. I have chosen to present the relatively simple and intuitive proof that uses the extraneous assumption that preferences are

increasing and strictly increasing along the diagonal, so that U is defined along the diagonal by $U((d, d, \dots, d)) = d$ and then extended to all of the commodity space. An in-class picture of this construction and a short discussion of what remains in the proof—namely, that this gives a representation (easy) and that the representation is continuous (harder)—helps enormously.

For students who want to see the proof without the extraneous assumption, all the details are supplied between Problem 2.14 and its solution in the *Student's Guide*.

The second hard point is all the notation surrounding separability. A concrete example sketched in class may help.

I've put separability, quasi-linearity, and homotheticity after the discussion of indifference curves (and monotonicity, continuity, and convexity) on two grounds:

1. Some instructors may wish to skip these three structural properties.
2. It may make more pedagogical sense to hold off on them until the consumer's demand problem is introduced in Chapter 3. Discussions of these three properties are helped if students can see how these assumptions might be used.

If you are going to cover these properties, I strongly suggest Problem 2.13, which discusses the consequences of quasi-linearity and homotheticity for indifference curves in R_+^2 . You might also ask students to ponder what it would take to reconstruct all the indifference curves for a consumer with utility function of the form $U((x_1, x_2)) = u_1(x_1) + u_2(x_2)$, although it must be noted that additive separability for $k = 2$ isn't axiomatized anywhere in this chapter.

Notes on the Problems

Problem 2.1 is very helpful and should be assigned. (The solution is in the *Student's Guide*.) Depending on whether you/your students have the "corrected" printing of the book or not, it may have only one part or two. If your copy of the book does not have a part b, please consult the Errata posted in the *Student's Guide*; I believe the point that "for all" is *not* strictly stronger than "for some" is worth making, and checking these properties against lexicographic preferences makes the point very nicely.

Problem 2.8(a) can be difficult, and 2.8(b) is virtually impossible. (Note that I don't give a solution of 2.8(b) in the *Guide* or here, although in the "solution" provided here, I give references to where a solution can be found.) As noted above, I think Problem 2.13 should be on the menu for in-class discussion, if quasi-linearity and homotheticity are going to be discussed at all. There is enough in the way of hints for mathematically sophisticated students to do Problem 2.14 (a fairly general proof of Debreu's theorem), but even mathematically sophisticated students may need some encouragement to persevere through all the steps. For students with the requisite background in measure theory, Problem 2.15 is quite interesting.

I assigned Problem 2.10 this year (Autumn 2013), and it caused some consternation among the students. See the commentary on this problem given when I get to its solution(s), following.

Solutions to Non-Starred Problems

- 2.2. Consider the preferences on R_+ created by the following utility function:

$$u(x) = \begin{cases} x, & \text{for } x < 1, \text{ and} \\ -x - 1, & \text{for } x \geq 1. \end{cases}$$

It is clear that for any convex set A that contains an open neighborhood of $x = 1$ (for instance, the interval $[0, 2]$), $c_u(A) = \emptyset$. The harder part is to show that these preferences are strictly convex, which amounts to showing that u is strictly quasi-concave:

Take x and y such that $x \succeq y$ (or, equivalently, such that $u(x) \geq u(y)$ and $x \neq y$). Since $x \neq y$ implies $u(x) \neq u(y)$ (the function u is one-to-one), we can assume that $u(x) > u(y)$. Now consider three cases:

- If $x, y < 1$, then $u(x) = x > u(y) = y$, and $u(ax + (1 - a)y) = ax + (1 - a)y > u(y)$.
- If $x, y \geq 1$, then $u(x) = -x - 1$ and $u(y) = -y - 1$, hence $u(ax + (1 - a)y) = -(ax + (1 - a)y) - 1 > u(y)$ (assuming $-x - 1 > -y - 1$).
- If one of x and y is strictly less than 1, and the other is greater or equal to 1, then $x < 1$ and $u(x) = x$, while $y \geq 1$ and $u(y) = -1 - y$. For any $a \in (0, 1)$, if $ax + (1 - a)y < 1$, then $u(ax + (1 - a)y) > 0 > u(y)$. And if $ax + (1 - a)y \geq 1$, then $u(ax + (1 - a)y) = -(ax + (1 - a)y) - 1 > -y - 1 = u(y)$. Done.

- 2.4. Suppose preferences $\succeq$ are convex. Let y and z be elements of $\text{NWT}(x)$, and let $a \in [0, 1]$. Note that $y \succeq x$ and $z \succeq x$. By completeness, either $y \succeq z$ or $z \succeq y$; without loss of generality, assume it is $y \succeq z$. Then by convexity of preferences, $ay + (1 - a)z \succeq z$, hence by transitivity of $\succeq$, $ay + (1 - a)z \succeq x$ and $ay + (1 - a)z \in \text{NWT}(x)$.

Suppose $\text{NWT}(x)$ is convex for all x . Take any pair x and y such that $x \succeq y$ and take any $a \in [0, 1]$. We know that $y \succeq y$ by completeness, hence x and y are both elements of $\text{NWT}(y)$. This set is convex by assumption, so $ax + (1 - a)y \in \text{NWT}(y)$, which means $ax + (1 - a)y \succeq y$, which proves that preferences are convex.

- 2.7. On grounds that a picture is worth 1000 words, Figure M2.1 is worth 2000.

- 2.8. (a) There are a number of ways to do this problem. For showing convexity of the two preferences, you can either show directly that the NWT sets are all convex (a bit tiresome in the case of U_2), or you can enlist Proposition 2.7 and show that $U_i(x_1, x_2)$ in each case is a quasi-concave function. The latter route can be accomplished by using the following two results:

Let g be a quasi-concave function defined on $C \subset R^n$ and let f be a nondecreasing function on $D \subset R$, containing the range of g . Then the composite function $U = f(g(x))$ is also quasi-concave, and



(a) An example with a bliss point

(b) An example with locally insatiability

Figure M2.1. Problem 2.7. In each panel, you see three points, one of which is greater in both components than the second, which is greater in both components than the third. The first and third are connected by a line segment. The first point plays the role of x , the second is x' , and the third is x'' . The line segment is the path ϕ from x and x'' ; let $\phi(t) = (1 - t)x + tx''$. In both panels, dashed curves are indifference curves. In panel a, think of x' as a local bliss point; the segment joining x and x'' then never hits a point as good as x' . In panel b we have an example with an indifference curve that is not unimodal; the reason for panel b is that this is an example in which preferences are everywhere locally insatiable.

A differentiable function f defined on an open convex set $C \subset \mathbb{R}^n$ is quasi-concave if and only if, for all $x, y \in C$, $f(y) \geq f(x)$ implies $(y - x)\nabla f(x) \geq 0$.

In the case of U_1 , it is easy to see that we cannot get a concave representation of these preferences because $(3, 3) \succ (2, 2)$ according to U_1 , while $(2.5, 2.5) \sim (2, 2)$; if we had a concave representation u , $(3, 3) \succ (2, 2)$ would imply $u(3, 3) > u(2, 2)$ hence $u(2.5, 2.5) \geq .5u(3, 3) + .5u(2, 2) > u(2, 2)$, and thus $(2.5, 2.5) \succ (2, 2)$. By the same example, U_1 gives preferences that are not semi-strictly convex. But the preferences given by U_1 are continuous; this is easily observed from the fact that U_1 is a continuous function.

For U_2 , seeing that we cannot get a concave representation is more difficult. If you know that every concave function is continuous on the interior of its domain of definition, you can note that these preferences are not continuous (demonstrated below). For a direct argument: (Draw pictures to accompany this argument.) Suppose that u were a concave representation. Note that u must equal some constant c for all points (x_1, x_2) such that $x_1 x_2 = 4$ and $x_1 \geq x_2$, and it must be some larger constant c' for all points (x_1, x_2) such that $x_1 x_2 = 4$ and $x_1 < x_2$. For $n = 1, 2, \dots$, let $x_1^n = y_2^n = 2 + (1/n)$, and let $x_2^n = y_1^n = 4/(2 + (1/n))$. That is, x^n and y^n are pairs of points symmetrically located (symmetric around the line $x_1 = x_2$) on the curve $x_1 x_2 = 4$ which (together) are getting closer and closer to the point $(2, 2)$. Let $z^n = .5(x^n + y^n)$; i.e., z^n is the midpoint of the line segment joining x^n and y^n , and so z^n is on the line $x_1 = x_2$. By concavity of u , $u(z^n) \geq .5(u(x^n) + u(y^n)) = .5(c + c')$. Now write the point $(2, 2)$ as a convex combination of $(0, 0)$ and z^n ; as $(2, 2) = a_n z^n + (1 - a_n)(0, 0)$, say. Since

z^n is approaching $(2, 2)$ as $n \rightarrow \infty$, $a_n \rightarrow 1$. By concavity of preferences again, $u((2, 2)) \geq a_n u(z^n) + (1 - a_n)u(0, 0) \geq a_n(.5)(c + c') + (1 - a_n)u((0, 0))$, where we have used the previously derived inequality for $u(z^n)$. As $n \rightarrow \infty$, $a_n \rightarrow 1$, so we get $u((2, 2)) \geq .5(c + c') > c$, which is a contradiction to the fact that $u((2, 2)) = c$. (Now try constructing an argument for the case in which U_2 is defined to be 5 at the point $(2, 2)$.)

The preferences given by U_2 are not continuous. (N.B., it is *not* enough to show that U_2 is not a continuous function. Continuous preferences can certainly be represented by discontinuous functions.) This is easy to see from the definition of continuous preferences that involves sequences of bundles: If we take z^n as above, then $z^n \succeq (1, 4)$ for all n , but the limit in n of z^n is $(2, 2)$, and it is not the case that $(2, 2) \succeq (1, 4)$. Finally, the preferences given by U_2 are semi-strictly convex. This involves a tedious exploration of numerous cases and is omitted.

(b) This is extremely complex, so I will only give a couple of comments and then references. Essentially, this example is constructed so that, within a triangle, indifference curves are straight lines but not parallel straight lines. Obviously parallel straight line indifference curves on a bounded region can be represented by a concave function; viz., by the linear function implicit in any one. But, it turns out, that if indifference curves are straight lines but not parallel, no concave representation is possible. See Y. Kannai, "Concavifiability and Constructions of Concave Utility Functions", *Journal of Mathematical Economics* (1977) 4:1–56. This, however, is more complex than necessary, because it deals with concavifiability in general. A somewhat informal argument (which can be made formal without much difficulty) for the case of straight-line indifference curves that are not parallel is given in R. Aumann, "Values of Markets with a Continuum of Traders," *Econometrica* (1975), 43:611–46.

■ 2.10. In the *Notes on the Problems* given previously, I remarked that this problem was the cause of some consternation among the students in my 2013 edition of the course. Some of the students were simply stuck; for those, I suggested that they first try to describe the shape of u in the special case of $k = 1$ and $\succeq \equiv \geq$. Also, I will give two proofs here. The first was the proof I put in this *Manual* originally. It is nice and slick, but it is too slick for some of my students: They can follow it step by step (they say), but they don't understand what is going on. So I've added a second proof, which is less elegant but easier to "see." And, inspired by the second proof, if you are looking for a nice take-home problem, challenge your students to "complete" the problem by finding necessary and sufficient conditions for the function u to be continuous at a point x . (Answer, u is continuous at x if and only if (a) x is not indifferent to any point $q \in Q_+^k$ or (b) x is not *locally immiserable*, where x is *not* locally immiserable if there is some (relatively) open neighborhood of x such that $x' \succeq x$ for every x' in the neighborhood. That is, x is locally immiserable if, within every (relatively) open neighborhood of x , there is some x' such that $x \succ x'$.)

First proof: Suppose that $\succeq$ is continuous and U is the function that represents $\succeq$ that

is produced in Propositions 1.12 and 1.15. For each $y \in X$, let $Q^*(y) = \{q^i \in Q_+^k : y \succeq q^i\}$, so that

$$u(y) = \sum_{i: q^i \in Q^*(y)} \frac{1}{2^i},$$

Suppose $\{x_n\}$ is a sequence from X with limit x . Let

$$Q_n^* = \bigcup_{m \geq n} Q^*(x_m) = \{q^i \in Q_+^k : x^m \succeq q^i \text{ for some } m \geq n\} \quad \text{and} \quad u_n = \sum_{i: q^i \in Q_n^*} \frac{1}{2^i}.$$

Then $u(x_n) \leq u_n$ since $Q^*(x_n) \subseteq Q_n^*$ and all the summands are positive. Therefore, $\limsup u(x_n) \leq \limsup u_n$, and we are done if we show that $\limsup u_n \leq u(x)$. For this last step, note that the sets Q_n^* are decreasing (in terms of set inclusion), and hence $\lim u_n$ exists and equals

$$\sum_{i: q^i \in \bigcap_n Q_n^*} \frac{1}{2^i}.$$

But $q^i \in \bigcap_n Q_n^*$ means that $x_n \succeq q^i$ for infinitely many n . That is, there is an infinite subsequence n' such that $x^{n'} \succeq q^i$ for all n' in the subsequence. Therefore, by continuity of $\succeq$, $x \succeq q^i$, hence $q^i \in Q^*(x)$. That is, $\bigcap_n Q_n^* \subseteq Q^*(x)$. Since all the summands are positive,

$$\lim u_n = \sum_{i: q^i \in \bigcap_n Q_n^*} \frac{1}{2^i} \leq \sum_{i: q^i \in Q^*(x)} \frac{1}{2^i} = u(x).$$

Second proof: First recall what it means for u to be upper semi-continuous. For all x and all sequences $\{x^n\}$ with limit x , it must be that $\limsup u(x^n) \leq u(x)$. Or, in other words, for every x , every sequence $\{x^n\}$ with limit x , and every $\epsilon > 0$, there is some N such that, for all $n \geq N$, $u(x^n) \leq u(x) + \epsilon$. Or, equivalently, for every x and $\epsilon > 0$, we can produce an open neighborhood of x (perhaps relatively open, if x is on the boundary of R_+^k) such that, for all x' in this neighborhood, $u(x') \leq u(x) + \epsilon$.

So, fix some $x \in R_+^k$ and some $\epsilon > 0$. Let M be such that $1/2^M < \epsilon$. Let $Q^*(x) = \{q \in Q_+^k : x \succeq q\}$ and let $\hat{Q}(x) = Q \setminus Q^*(x)$. That is, $Q^*(x)$ are all the rational-component points that are no better than x , and $\hat{Q}(x)$ are all the rational-component points that are strictly better than x . Of course, by definition,

$$u(x) = \sum_{i: q^i \in Q^*(x)} \frac{1}{2^i}.$$

Now consider the set $\hat{Q}^M(x) = \{q^i; i = 1, \dots, M\} \cap \hat{Q}(x)$. In words, $\hat{Q}^M(x)$ is the set of rational-component points whose index in the enumeration is less or equal to M and that are strictly preferred to x . Of course, this is a finite set and, equally of course, x is not in this set. Since preferences are continuous and x is strictly worse than each $q \in \hat{Q}^M(x)$, for each such q we can find an open neighborhood of x such that all points in the neighborhood are strictly worse than this q . Since there are finitely many such q , the intersection of these open neighborhoods is an open neighborhood of x , such that every x' in the intersection is strictly worse than each $q \in \hat{Q}^M(x)$.

But, then, I assert that for every x' in this intersection-of-finitely-many-open-neighborhoods, $u(x') \leq u(x) + \epsilon$. The reason is that, if we compare the sums that define $u(x')$ and $u(x)$ (sums of strictly positive terms), the only terms that can appear in the sum that defines $u(x')$ and not appear in the sum that defines $u(x)$ are terms $1/2^i$ for $i > M$. And these terms can sum to no more than $1/2^M$, which is (by construction) $< \epsilon$. ■

■ 2.12. The direction *If preferences have a continuous and homogeneous representation, then they are continuous and homothetic* is very easy. Continuity of the representation implies continuity of the preferences. As for homotheticity and homogeneity, suppose $x \succeq y$. Then $U(x) \geq U(y)$. Take any $\lambda \geq 0$, and we know that $\lambda U(x) \geq \lambda U(y)$. But $\lambda U(x) = U(\lambda x)$ and similarly for y , so $U(\lambda x) \geq U(\lambda y)$ which implies $\lambda x \succeq \lambda y$. Done.

If we add the assumption that preferences are monotone and strictly monotone along the diagonal, proving the existence of a homogeneous representation follows the proof of Debreu's theorem given in the text: We know that for each $x \in X$, there is some $d(x) \in R_+$ such that $x \sim d(x)e$, where $e = (1, 1, \dots, 1) \in R_+^k$. Define $U(x) = d(x)$. Still following the proof of Debreu's theorem from the text, we know that U represents $\succeq$. So it remains to show that U is homogeneous. Take any x and $\lambda \geq 0$. By homotheticity, since $x \sim d(x)e$, we know that $\lambda x \sim \lambda d(x)e$, thus $d(\lambda x) = \lambda d(x)$, and thus $U(\lambda x) = \lambda U(x)$. And continuity of the representation follows immediately from the proof given in the text.

Things are a bit harder if we don't make any monotonicity assumptions. I'll sketch the argument. First we argue that for each $x \neq 0$ from R_+^k , either $x \sim 0$, in which case $\lambda x \sim 0$ for all $\lambda \geq 0$, or $x \succ 0$, in which case $\lambda x \succ 0$ for all $\lambda > 0$, or $0 \succ x$, in which case $0 \succ \lambda x$ for all $\lambda > 0$. Note in the latter two that we are extending strict preference for strictly positive scalars λ ; the idea is that if, in the second case, say, $0 \succeq \lambda x$, then $0 = (1/\lambda)0 \succeq (1/\lambda)\lambda x = x$, contradicting $x \succ 0$.

Suppose $x \succ 0$ and $y \succ 0$. Then there exists some $\lambda > 0$ such that $x \sim \lambda y$: To see this, note that as $\alpha \rightarrow 0$, $\alpha x \rightarrow 0$, so by continuity, $y \succ \alpha x$ for some $\alpha > 0$. By Lemma 2.10, there is then β strictly between zero and one such that $\beta y \sim \alpha x$, and $\lambda = \beta/\alpha$ by homotheticity.

Similarly, if $0 \succ x$ and $0 \succ y$, there is some $\lambda > 0$ such that $x \sim \lambda y$.

Pick any $\bar{x}$ such that $\bar{x} \succ 0$, if one such exists, and pick any $\underline{x}$ such that $0 \succ \underline{x}$, if

one such exists. Define $U(\lambda\bar{x}) = \lambda$ and $U(\lambda\underline{x}) = -\lambda$, both for all $\lambda > 0$. Use the previous two paragraphs to extend U to all of R_+^k , where (of course) we set $U(x) = 0$ if $x \sim 0$. We have to show that U gives a representation of $\succeq$, which is relatively simple: U gives positive values to x such that $x \succ 0$ and negative values when $0 \succ x$, and the argument from the proof of Debreu's theorem shows that U works (gives a representation) within each class separately. And we have to show that U is homogeneous, which essentially follows the argument given four paragraphs ago.

We have to work a bit, to show that this representation is continuous. Suppose $\{x^n\}$ has limit $x \succ 0$. Then for all sufficiently large n , $x^n \succ 0$. We know that (for all large n) $x^n \sim u(x^n)\bar{x}$ and $x \sim u(x)\bar{x}$; suppose $\limsup u(x^n) > u(x)$. Write α for the limit superior, λ for $u(x)$, and $\beta = (\alpha + \lambda)/2$. Of course $\alpha > \beta > \lambda$, and so $\alpha\bar{x} \succ \beta\bar{x} \succ \lambda\bar{x}$. Hence $\alpha\bar{x} \succ \beta\bar{x} \succ x$. Since $x^n \rightarrow x$, for all sufficiently large n , $\beta\bar{x} \succ x^n$. But $x^n \sim u(x^n)\bar{x}$, and for arbitrarily large n , $u(x^n) > \beta$, so for those n , $x^n \succ \beta\bar{x}$, a contradiction. A similar argument shows that $\liminf u(x^n)$ must equal $u(x)$.

And a similar argument shows that we will have continuity at points x such that $0 \succ x$.

This leaves the case $x \sim 0$. In this case $u(x) = 0$. Suppose that $\limsup u(x^n) = \alpha > 0$. Then for arbitrarily large n , $x^n \sim u(x^n)\bar{x} \succ (\alpha/2)\bar{x} \succ 0$. This is, once again, a contradiction to $\lim_n x^n = x \sim 0$, if preferences are continuous. And, similarly, one shows that $\liminf u(x^n)$ must be 0, if $x \sim 0$.

Note here that U is being defined "independently" on the pieces of R_+^k which are better and worse than the origin. That is, we could rescale U on one piece or the other, without affecting the result. To understand this point, note that any function f from R to R that has $f(0) = 0$ and is piecewise linear on $[0, \infty)$ and on $(-\infty, 0]$ is homogeneous, even if it is kinked at the origin.

■ 2.13. Suppose we have a single indifference curve and we know that preferences are quasi-linear in the second good. Then all indifference curves are a vertical translation of the one you have. This doesn't (necessarily) tell you everything about the consumer's preferences; if the indifference curve you have hits the horizontal axis at, say $(x^*, 0)$, then you only have indifference curves up to the level x^* of the first good. See Figure M2.2.

Note that we have drawn our original indifference curve so that it hits the vertical axis. Condition c in Proposition 2.16 guarantees that this must happen: Take any (x, m) along the indifference curve. There is some m' and m'' such that $(x, m') \sim (0, m'')$. Because preferences are strictly monotone, $m'' > m'$. But then $(0, m + m'' - m') \sim (x, m)$ by condition b.

Note also that we have drawn the original indifference curve so that it hits the horizontal axis. This is not necessarily guaranteed. In fact, if $\lim_{x \rightarrow \infty} u(x) = \infty$ in the representation, then every indifference curve will hit the horizontal axis, while if $\lim_{x \rightarrow \infty} u(x) = k < \infty$, then the indifference curve through $(0, k)$ will asymptote to the horizontal axis,

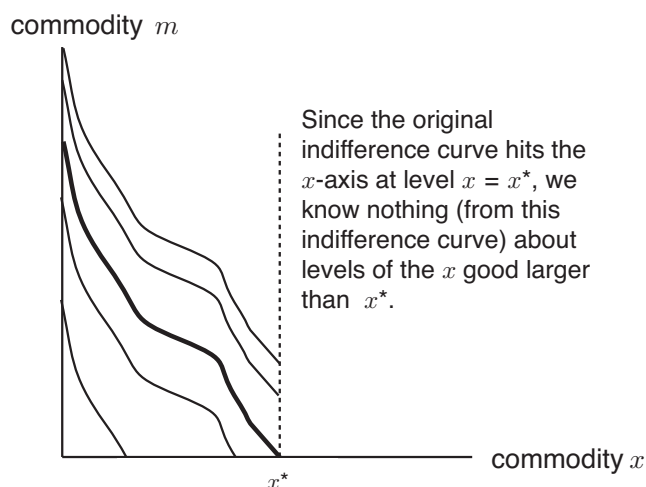


Figure M2.2. Indifference curves for quasi-linear preferences. When preferences are quasi-linear in the second good, indifference curves are just vertical translates of one another. Given one indifference curve, shown here in bold, we know all the others for the range of x covered by the first indifference curve.

indifference curves through $(0, m)$ for $m < k$ will hit the horizontal axis, and indifference curves through $(0, m)$ for $m > k$ will asymptote above the horizontal axis. (Prove these things if they aren't obvious to you.)

If instead we know that preferences are homothetic, and if the indifference curve we are given is anything other than the “curve” consisting of the origin, we know everything about the consumer's preferences: Every indifference curve is the radial projection of this one. Put another way, take any point x . By continuity, we know that the ray from the origin through x must pass through the one indifference curve. Let $i(x)$ be this “projection” of x onto the indifference curve, and let $U(x)$ be the norm of x divided by the norm of $i(x)$. Then it is easy to see that $U(x)$ gives a numerical representation of preferences. See Figure M2.3.

Note that we have drawn the original indifference curve in this case as hitting both axes. If preferences are strictly monotone, continuous, and homothetic, this must happen: By strict monotonicity $(1, 1) \succ (0, 1) \succ (0, 0)$, and hence by continuity there is some $d > 0$ such that $(d, d) \sim (0, 1)$. But then if the indifference curve passes through the diagonal at (d', d') , by homotheticity we know that $(d', d') \sim (0, d'/d)$; i.e., the indifference curve hits the vertical axis at d'/d .

■ 2.15. In this problem $u(x) = \mu(\text{NBT}(x))$. To know that this definition makes sense (before trying to prove anything), we need to know that the set $\text{NBT}(x)$ is measurable and $u(x)$ is finite. We are assuming that μ is a nonnegative measure and $\mu(R_+^2) < \infty$, so finiteness is no problem. And because $\succeq$ is continuous, $\text{NBT}(x)$ is a closed set, and all closed sets are (Borel) measurable.

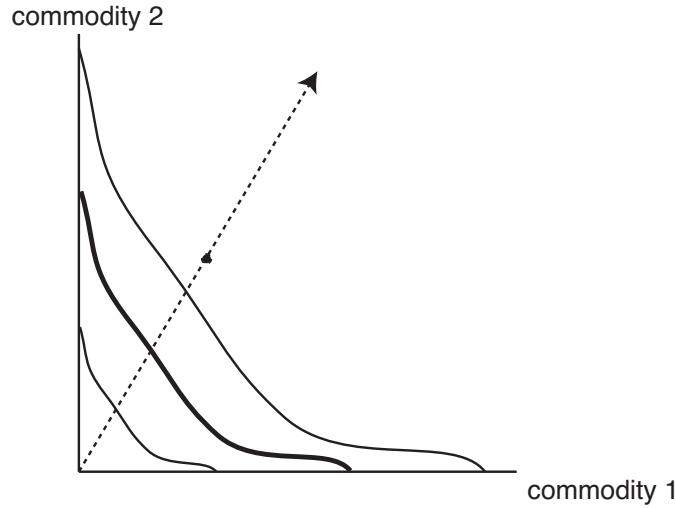


Figure M2.3. Indifference curves for homothetic preferences. Every indifference curve is a radial projection of every other indifference curve. Given the original indifference curve, depicted in bold, we have depicted here a second that is 50% further out from the origin and a third that is half-way in. Note the point marked by the dot. On the ray from the origin through it, the ratio of the distance from the origin to the point to the distance from the origin to the original indifference curve is 1.8. Hence we can construct a homogeneous utility function for these preferences with $U(x) = 1$ for every point x along the original indifference curve, and for this utility function U at the heavily marked point would be 1.8.

First, we show that u represents $\succeq$.

As long as $\succeq$ is complete and transitive, Proposition 1.17 holds: $\text{NBT}(x) \subseteq \text{NBT}(y)$ if and only if $y \succeq x$. Hence, $y \succeq x$ implies $\text{NBT}(x) \subseteq \text{NBT}(y)$ implies $u(x) = \mu(\text{NBT}(x)) \leq \mu(\text{NBT}(y)) = u(y)$, since for nonnegative measures, the measure of a bigger set is always (weakly) bigger.

The converse half, $u(x) \geq u(y)$ implies $x \succeq y$, is where continuity comes in. I prove the contrapositive: If $y \succ x$, then $u(y) > u(x)$: Suppose $y \succ x$. Then per the proof of Proposition 1.15 in the text, we know (since $\succeq$ is continuous) that there is some $z \in R_+^2$ such that $y \succ z \succ x$. Moreover, by the definition of continuity, we know there is an open ball of positive radius, call it Z , such that $z \in Z$ and, for all $z' \in Z$, $y \succ z' \succ x$. But this means that $Z \subseteq \text{NBT}(y)$ and $Z \cap \text{NBT}(x) = \emptyset$. Therefore, $u(y) - u(x) = \mu(\text{NBT}(y)) - \mu(\text{NBT}(x)) = \mu(\text{NBT}(y) \setminus \text{NBT}(x)) \geq \mu(Z)$ (because $Z \subseteq \text{NBT}(y) \setminus \text{NBT}(x)$), and since Z is an open ball of positive radius, it has positive Lebesgue measure, hence positive μ measure. Therefore, $u(y) - u(x) \geq \mu(Z) > 0$, or $u(y) > u(x)$.

But, no, u is not necessarily continuous.

Suppose that $\succeq$ is the preference relation given by $v((x_1, x_2)) = \max\{x_1 + x_2, 2\}$. This utility function v is continuous, so we know that $\succeq$ is continuous. And it is a numer-

ical representation of the preferences it engenders, so they are complete and transitive. Now let $x^n = (1 - \frac{1}{n}, 1 - \frac{1}{n})$. This has limit $(1, 1)$, of course. But $u(x^n)$ does not approach $u((1, 1))$: $u(x^n) = \mu(\{x = (x_1, x_2) \in R_+^2 : x_1 + x_2 \leq 2 - 2/n\})$, while $u((1, 1)) = \mu(R_+^2)$. The difference $u((1, 1)) - u(x^n) \geq \mu(\{x = (x_1, x_2) \in R_+^2 : x_1 + x_2 \geq 2\})$, and as $\{x = (x_1, x_2) \in R_+^2 : x_1 + x_2 \geq 2\}$ is a set of strictly positive Lebesgue measure, it has strictly positive μ measure.

For the rest of the problem, it helps to be able to deal with sequences that converge to x monotonically, either from "above" or from "below." It isn't necessary to do this, but for some initial investment in notation, it makes life a lot easier later on. So I'll make the necessary investment. Imagine that we have some sequence $\{x_n\}$ in R_+^2 that converges to some point x . Re-label x as x^∞ and, for this sequence, define

$$\bar{x}^n = \text{preference max}\{x^m; m = n, n+1, \dots, \infty\} \text{ and}$$

$$\underline{x}^n = \text{preference min}\{x^m; m = n, n+1, \dots, \infty\}.$$

To explain, $\bar{x}^n$ is that x^m for $m \geq n$ such that $x^m \succeq x^\ell$ for all $\ell \geq n$, including $\ell = \infty$, and symmetrically for $\underline{x}^n$. (If there is a tie for this honor, then take, say, the x^m that satisfies this condition and has the smallest index.) How do I know that this preference-maximum exists? There are two cases to consider:

Suppose $x^\infty \succeq x^m$ for every $m = n, n+1, \dots$. Then if $x^m \sim x^\infty$ for some finite m , $\bar{x}^n$ will be that x^m of smallest index such that this is true. And if $x^\infty \succ x^m$ for all finite m , then $\bar{x}^n$ will be x^∞ .

The second case is there $x^m \succ x^\infty$ for some finite $m \geq n$. Let M be the smallest index (greater than or equal to n) such that this is true. Since the sequence converges to x and $\succeq$ is continuous, there is some large $N > M$ such that $x^M \succ x^\ell$ for all $\ell > N$. But then the candidates for $\bar{x}^n$ are members of the finite set $\{x^M, x^{M+1}, x^N\}$. There is a preference maximizing member of this set, and either that (or something of smaller index that is indifferent to it) is our $\bar{x}^n$.

Existence of $\underline{x}^n$ for each n is proved similarly.

But since $\bar{x}^n$ is x^ℓ for some $\ell \geq n$ (including, possibly, $\ell = \infty$), and similarly for $\underline{x}^n$, and since $\{x^n\}$ converges to x^∞ , we know that $\lim_n \bar{x}^n = \lim_n \underline{x}^n = x^\infty$. Let me spell this out: Since $x^n \rightarrow x$, for all $\epsilon > 0$, there is some N such that, for all $n \geq N$, $\|x^n - x\| < \epsilon$. But then, for this very N , if $n \geq N$, since $\bar{x}^n = x^\ell$ for $\ell \geq n \geq N$, and $\underline{x}^n = x^j$ for some $j \geq n \geq N$, $\|\bar{x}^n - x^\infty\|$ and $\|\underline{x}^n - x^\infty\|$ are both $< \epsilon$.

Moreover, convergence is monotonic (in terms of preference), monotonic non-preference-increasing for the first sequence and monotonic non-preference-decreasing for the second or, in other words

$$\bar{x}^1 \succeq \bar{x}^2 \dots \succeq x^\infty \succeq \dots \succeq \underline{x}^2 \succeq \underline{x}^1.$$

Finally, for each n , $\bar{x}^n \succeq x^n \succeq \underline{x}^n$, hence $u(\bar{x}^n) \geq u(x^n) \geq u(\underline{x}^n)$ for each n .

That gives me the tools I need to proceed. The first further step is:

u is, in fact, upper semi-continuous

Suppose that $\{x_n\}$ approaches x . Then so does $\{\bar{x}^n\}$, and since $u(\bar{x}^n) \geq u(x^n)$ for each n , if we show that $\lim_n u(\bar{x}^n) = u(x)$, we will know that $\limsup_n u(x^n) \leq u(x)$, which is upper semi-continuity of u .

But $u(\bar{x}^n) = \mu(\text{NBT}(\bar{x}^n))$. Since $\bar{x}^n \succeq \bar{x}^{n+1}$ for each n , $\text{NBT}(\bar{x}^n) \supseteq \text{NBT}(\bar{x}^{n+1})$, and so by the continuity of measure, $\lim_n u(\bar{x}^n) = \lim_n \mu(\text{NBT}(\bar{x}^n)) = \mu(\cap_n \text{NBT}(\bar{x}^n))$, where, to be clear, by $\cap_n \text{NBT}(\bar{x}^n)$ we mean the nonincreasing intersection of the $\text{NBT}(\bar{x}^n)$, for $n = 1, 2, \dots$; that is, not including $\text{NBT}(x)$. But I assert that $\cap_n \text{NBT}(\bar{x}^n) = \text{NBT}(x)$. Since each $\bar{x}^n \succeq x$, it is clear that $\text{NBT}(x) \subseteq \text{NBT}(\bar{x}^n)$ for each n , and hence $\text{NBT}(x) \subseteq \cap_n \text{NBT}(\bar{x}^n)$. Now suppose $y \in \cap_n \text{NBT}(\bar{x}^n)$, which is to say that $y \in \text{NBT}(\bar{x}^n)$ (since the sets are nonincreasing), which means $\bar{x}^n \succeq y$ for all n . Since $\succeq$ is continuous and $\bar{x}^n \rightarrow x$, this implies $x \succeq y$, or $y \in \text{NBT}(x)$. Done.

Therefore, the problem is one of lack of lower semi-continuity.

We see that in the counterexample presented earlier, but to be more systematic about it: Let $I(x) = \{y \in R_+^2 : y \sim x\}$, and let $\text{SWT}(x) = \{y \in X : x \succ y\}$. That is $I(x)$ is the indifference set of x , while $\text{SWT}(x)$ is the set of bundles strictly worse than x . We know that $\text{SWT}(x)$ is always open (it is a partial characterization of continuous preferences), while $I(x)$ is a closed set, because it is the intersection of the closed set $\text{NBT}(x)$ and the closed set $\text{NWT}(x) = \{y \in R_+^2 : y \succeq x\}$.

Take any sequence $\{x_n\}$ that approaches a given x . In some cases (where, for all sufficiently large n , $x^n \succeq x$), $\underline{x}^n \sim x$ for all sufficiently large n . In other cases (where $x^n \prec x$ infinitely often in n), $x \succ \underline{x}^n$ for all n . In the first of these two cases, $u(\underline{x}^n) = u(x)$ for all sufficiently large n , and there is no problem.

But if $x \succ \underline{x}^n$ for all n , then $\lim_n u(\underline{x}^n) = \lim_n \mu(\text{NBT}(\underline{x}^n))$ = by continuity of measure, since the sets are nondecreasing in size $\mu(\cup_n \text{NBT}(\underline{x}^n))$. And $\cup_n \text{NBT}(\underline{x}^n)$ in these cases is $\text{SWT}(x)$, not $\text{NBT}(x)$. To see that $\cup_n \text{NBT}(\underline{x}^n) = \text{SWT}(x)$, note first that if $y \succeq x$, then since $x \succ \underline{x}^n$ for all n , $y \notin \text{NBT}(\underline{x}^n)$, hence it cannot be in the union of those sets. But if $x \succ y$ (so that $y \in \text{SWT}(x)$), then by continuity (since $\underline{x}^n \rightarrow x$), $\underline{x}^n \succ y$ for large enough n , hence $y \in \text{NBT}(\underline{x}^n)$ for this n , and hence y is in the union of the $\text{NBT}(\underline{x}^n)$ sets.

Now $u(x) = \mu(\text{NBT}(x)) = \mu(\text{SWT}(x)) + \mu(I(x))$, since $\text{NBT}(x)$ is the disjoint union of $\text{SWT}(x)$ and $I(x)$. Putting all this together, we have proved:

Suppose $\{x^n\}$ has limit x and suppose that $\mu(I(x)) = 0$. Then $\lim_n u(\underline{x}^n) = \mu(\text{SWT}(x)) = \mu(\text{NBT}(x))$ (since $\mu(I(x)) = 0$) $= u(x)$, hence u is lower semi-continuous at x . And, combining with earlier results, u is continuous at x .

Therefore, for u to be continuous, it is sufficient that $\mu(I(x)) = 0$ for all x .

(In fact, this is virtually necessary as well. The only points whose indifference curves can have positive measure and still have u continuous would be x that are no better than every other $y \in R_+^k$. But I won't take you through the proof of that.)

We could stop here and say, for u to be continuous, every indifference set must have zero μ (hence zero Lebesgue) measure. But indifference sets having zero measure is not as economically meaningful as one might like. Can we recast this in terms of properties of $\succeq$ that are closer to "economics?"

The obvious conjecture at this point is

If $\succeq$ is locally insatiable and continuous, then its indifference sets all have zero Lebesgue measure, hence u is continuous.

And that may well be true, but I can neither prove it nor offer a counterexample. To indicate the difficulties, consider the following conjecture:

Suppose $\succeq$ is continuous and has the property that, in every neighborhood of every point x there is some x' that is not indifferent to x . (In other words, $I(x)$ contains no open neighborhoods of any strictly positive radius.) Then each $I(x)$ has zero Lebesgue measure, and u is continuous.

Compare this with the previous conjecture. Local insatiability implies this, because it says in every neighborhood of x there is some point strictly better than x . This says only that there is some point strictly better or strictly worse than x .

The second conjecture is false. Here's my counterexample. Take all the points $q \in R_+^2$ that have rational coordinates. Enumerate these points as $\{q_1, q_2, q_3, \dots\}$. Let Q_n be an open ball around q_n whose area is $0.01/2^n$. (That is, its radius is $0.1/(2^{n/2}\pi^{1/2})$). Let Q be the union of all these Q_n . Since each Q_n is open, so is Q . Clearly, the Lebesgue measure of Q is less than the sum of the areas of all those balls (less because they will overlap to some extent), and the sum of the areas of all the balls is 0.01. So Q is an open set, everywhere dense in R_+^2 (because it contains all points with rational coordinates), whose Lebesgue measure is less than 0.01. The complement of Q , which I'll denote by Q^C , is a closed set.

Now for each $x \in R_+^k$, define $v(x)$ to be the minimal distance from x to Q^C . I assert that v is a bounded and continuous function. For the upper bound, note that since the measure of Q is 0.01, there are points in Q^C in every "square" of the form $\{y \in R_+^k : (n, m) \leq y \leq (n+1, m+1) \text{ for integers } n \text{ and } m\}$, since each such square has Lebesgue measure 1. Any x , then, can be no more than $\sqrt{2}$ from some point in Q^C ; there is some point in x 's square that is in Q^C . As for continuity of u , note that the problem

$$\min \|x - y\|, \text{ for fixed } x \text{ and } y \in Q^C,$$

has a solution— Q^C is closed, and now that we know that this distance is bounded, we can restrict attention for any x to a bounded subset of Q^C . So, for each $x \in R_+^2$,

let $x^*(x)$ be the closest point to x in Q^C . (There may be more than one, in which case just select one.)

Now suppose $x^n \rightarrow x$. It is clear that $x^*(x^n)$ lives inside a bounded set, so along some convergent sequence converges to some y . Of course, since Q^C is closed, $y \in Q^C$. Since $v(x^n) = \|x^*(x^n) - x^n\|$, it follows by continuity of the norm that, along this subsequence, $\lim \|x^*(x^n) - x^n\| = \|y - x\|$. This implies that $v(x) \leq \liminf_n v(x^n)$. (Take a subsequence in n along which the limit infimum is attained, refine the subsequence so that the $x^*(x^n)$ converge, and...) And since $v(x) = \|x^*(x) - x\| = \lim_n \|x^*(x) - x^n\|$ and $v(x^n) \leq \|x^*(x) - x^n\|$ for each n , it is clear that $\lim \sum_n v(x^n) \leq v(x)$. That's continuity.

Note that, of course, $v(x) = 0$ for all $x \in Q^C$. And for $x \in Q$, $v(x) > 0$.

So let $\succeq$ be the preferences given by v . Since v is continuous, those preferences are continuous. Of course they are complete and transitive. Because Q is everywhere dense in R_2^k and $v(x) > 0$ if $x \in Q$, in any neighborhood of any point y in Q^C (so $v(y) = 0$), there is some point strictly preferred to y . Now take any point $x \in Q$. Each convex combination $\alpha x + (1 - \alpha)x^*(x)$ for $\alpha < 1$ is closer to Q^C than is x , so $v(\alpha x + (1 - \alpha)x^*(x)) < v(x)$ for $\alpha < 1$. So while x may be a local maximum of v , in every neighborhood, v is somewhere smaller.

So the $\succeq$ derived from v meets the conditions of my conjecture. But the indifference set Q^C has positive measure.

You can complain that while this indifference set has positive measure, it consists of all the worst points, so its u is in fact continuous. In which case, take $w(x) = -v(x)$, and we're there. (Now it is definitely the case that points in Q^C are all local preference maxima.)

To reiterate, it might be that the conjecture concerning local insatiability is true. I don't have a counterexample to that. But this construction indicates that proving that conjecture might be difficult.

There are more restrictive conditions I can give on $\succeq$ that guarantee that all indifference sets have zero Lebesgue measure. For instance:

Suppose $\succeq$ is strictly monotone and continuous. Then its indifference sets all have zero Lebesgue measure, hence u is continuous.

To see this, fix x . For each $y_1 \in R_+$, there can be no more than one y_2 such that $x \sim (y_1, y_2)$. Now apply Fubini: The measure of $I(x)$ is the integral over y^1 of the measures of "slices" of $I(x)$ (fixing the first coordinate), which is obviously zero, since the measure of a single point is always zero. I'm fairly certain that a similar argument can be given if $\succeq$ is only (!) strictly monotone for strict increases—I'm not expert enough to know about trying to use Fubini in radial coordinates (which is the obvious way to proceed), but I'm fairly sure I can manage an argument where I'll bound the outer measure of each $I(x)$ by an arbitrarily small number. Since this solution has gone on long enough, I'll leave it here.

When I assigned this problem in class, one student made a very interesting proposal: He said u becomes continuous if you remove from the integral (or from each $\text{NBT}(x)$) indifference sets of positive measure. But he provided no further details, and I'm not sure I know how the details would go; for one thing, do we know anything in general about the union of all indifference sets of positive measure. Is the union a measurable set? Do we know, for instance, that there can be no more than a countable number of such indifference sets? I haven't thought it through, so I don't know if it can be made to work. But if anyone wants to give more thought to this, it is an interesting idea to explore.